Do AI Explanations Improve Human Decisions or Merely Increase Perceived Trust?

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Abstract

This review examines whether AI explanations improve human decisions or mainly increase perceived trust. Twelve studies are synthesised across experiments, surveys, reviews, applied workflows, and conceptual analysis. The evidence is mixed. Explanations help when they are accurate, verifiable, and suited to the user's knowledge. They can also increase reliance without improving accuracy. Misleading explanations may cause immediate errors and lasting misinformation. The review argues that explainable AI should aim for appropriate reliance rather than maximum trust.

1 Introduction

Artificial intelligence increasingly supports decisions in healthcare, government, education, finance, and other high-stakes settings. In these contexts, full automation is often undesirable. AI systems can process large amounts of data, but they also make errors. Human decision-makers retain contextual knowledge and formal responsibility. The expected benefit therefore comes from collaboration rather than replacement.

Explainable artificial intelligence (XAI) is commonly proposed as the bridge between opaque models and responsible human oversight. An explanation may show important features, provide similar examples, state a confidence score, or describe why a recommendation was produced. The expected sequence is intuitive: explanation leads to understanding, understanding leads to calibrated trust, and calibrated trust leads to better decisions. The empirical literature shows that each link in this sequence can fail.

The central problem is that explanations have both an informational and a persuasive function. They may reveal useful evidence. They may also make a recommendation appear reasonable. These effects are not equivalent. A person may feel informed while remaining unable to identify an AI error. A clear explanation may increase confidence without improving final accuracy. A plausible but incorrect explanation may even teach the wrong reasoning strategy.

This review asks: Do AI explanations improve human decisions, or do they mainly increase perceived trust? Twelve selected articles are examined. They cover controlled XAI experiments, human-in-the-loop studies, trust surveys, applied assessment workflows, a meta-analysis, and conceptual work. The review focuses on five outcomes: decision accuracy, error detection, trust, reliance, and learning. It argues that explanations are useful only when they support critical evaluation. Explanations that merely increase perceived transparency can create an illusion of control.

2 Machine Learning Models and Explanation Methods

The studies use different models and explanation methods. This matters because explanation quality depends partly on the distance between the prediction and the explanation. Rule-based systems expose

their logic directly. Complex learned models often require a separate post-hoc explanation. This extra layer can make the explanation easier to read, but less faithful to the original prediction process.

2.1 Model complexity and traceability

Wang and Yin compare logistic regression with a multilayer neural network [11]. Logistic regression combines features in a relatively transparent form:

$$P(y = 1 \mid \mathbf{x}) = \sigma\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_j\right). \tag{1}$$

Each coefficient has a stable direction, so users can inspect how a feature affects the prediction. Neural networks capture nonlinear relationships and interactions, but their internal process is harder to trace. Wang and Yin find that explanations become less effective as model complexity increases. This suggests that an explanation of a complex model should not be treated as a complete account of its reasoning.

Janssen et al. compare machine learning with explicit business rules [6]. Rule-based recommendations were easier to trace, especially for experienced users. Yet transparency alone did not prevent all mistakes. The relevant lesson is that model accuracy and traceability must be considered together.

Generative systems create a further difficulty. Ersozlu et al. use a generative AI workflow to produce evidence-linked assessment feedback [5]. Albashrawi reviews GenAI applications across several decision domains [2]. These systems produce fluent text, but fluency is not proof of factual or logical reliability. Their outputs require independent verification.

2.2 Explanation formats

Feature-based explanations show which inputs influenced a prediction. They can help knowledgeable users assess whether the model relies on sensible factors. Wang and Yin report useful effects in familiar tasks, while Chen et al. find that feature-based explanations can increase overreliance without improving outcomes [11, 4]. A ranked list of features may therefore inform an expert but merely appear precise to a novice. Feature influence also describes model behaviour, not real-world causation.

Example-based explanations present similar cases. Chen et al. find that they supported intuition and stronger human–AI complementarity than feature-based explanations [4]. Counterfactual explanations show what would need to change for another outcome. They can be practical, but Wang and Yin find that they do not consistently improve calibrated trust [11]. Their usefulness depends on whether the proposed change is realistic and relevant.

Confidence scores communicate uncertainty. Zhang et al. show that case-specific confidence can improve trust calibration [12]. Yet confidence helps only when the estimate is calibrated and understood. A high but unreliable probability may strengthen overreliance rather than prevent it.

Post-hoc and natural-language explanations require particular caution. The prediction and explanation may be generated by separate processes. Cabitza et al. and Spitzer et al. show that correct advice can be paired with a misleading explanation, with negative effects on immediate accuracy and later learning [3, 10]. Peters adds that AI-generated reasons should not be interpreted as equivalent to human reasons or as direct access to an internal reasoning process [7].

The technical implication is limited but important. Explainability is not one uniform intervention. Feature contributions, examples, counterfactuals, confidence values, and generated text provide different evidence. They also create different risks. Their value must therefore be judged in relation to the model, the task, and the user's expertise.

3 Conceptual Distinctions

3.1 Trust, reliance, and decision quality

Trust is an attitude. It concerns the belief that a system is competent, dependable, or well intentioned. Reliance is behaviour. It concerns whether a person follows the system's advice. These concepts are related, but they are not interchangeable. A user may trust a system in general but reject one recommendation. A user may also follow advice because of time pressure without reporting strong trust.

The most useful standard is *appropriate reliance*. It requires accepting correct AI advice and rejecting incorrect AI advice. Let $A_i = 1$ when the AI advice is correct in case i, and let $R_i = 1$ when the user relies on it. A simple case-level measure is:

$$AR = \frac{1}{N} \sum_{i=1}^{N} \left[A_i R_i + (1 - A_i)(1 - R_i) \right]. \tag{2}$$

This measure rewards reliance when the AI is correct and non-reliance when it is wrong. It is more informative than an average trust score. It also makes clear why increasing reliance is not always desirable.

Decision quality should be measured separately. It may include final accuracy, error detection, consistency, speed, cognitive load, or learning after the AI is removed. Zhang et al. make this separation explicit. Their experiments show that confidence information can improve trust calibration, while better calibration does not necessarily improve joint performance. In their words, "trust calibration alone is not sufficient" [12]. Humans must also possess knowledge that complements the AI's errors.

3.2 Explanation quality

An explanation can be understandable without being faithful. It can be plausible without being relevant. Four qualities should be distinguished:

- **Understandability**: the user can interpret the explanation.
- **Plausibility**: the explanation appears reasonable.
- **Faithfulness**: the explanation reflects the model's actual basis.
- **Decision usefulness**: the explanation improves evaluation or action.

These dimensions often diverge. Peters argues that AI-generated reasons do not have the same regulative role as human reasons. A person can be held to a stated justification and may adjust later behaviour to remain consistent with it. A post-hoc AI explanation does not regulate the underlying model in this way [7]. Users may therefore apply social expectations of human reason-giving to a technical output that does not deserve them.

4 When Explanations Improve Decisions

The positive evidence is conditional rather than universal. Janssen et al. provide one of the clearest examples. They compared unaided decisions, machine-learning support, and business-rule support in government scenarios. Their experiment included 180 participants, divided by prior experience. Algorithmic support improved average performance. Transparent business rules produced more correct decisions than opaque machine learning. Experienced participants also performed better than inexperienced participants. Yet no group detected all algorithmic mistakes [6]. Explainability helped, but it did not remove automation risk.

Chen et al. show that explanation type matters. Their mixed-methods study compared feature-based and example-based explanations in two prediction tasks. Example-based explanations supported higher accuracy than feature-based explanations. They also helped participants combine AI information with

their own intuition. Feature-based explanations did not improve performance and increased overreliance. The authors identify three pathways through which users override AI: strong intuition about the outcome, reasoning about important features, and recognition of AI limitations [4]. Explanations were useful when they supported these forms of reasoning rather than replacing them.

Wang and Yin report a similarly conditional pattern. They compare feature importance, feature contribution, nearest-neighbour, and counterfactual explanations. The effects vary with domain familiarity and model complexity. When users had some knowledge of the task, feature-contribution explanations met more of the authors' criteria for effective XAI. When users lacked domain knowledge, none of the explanation types reliably improved understanding, uncertainty recognition, or calibrated trust. Effectiveness also declined for more complex models [11]. The result challenges the assumption that a human-friendly format is sufficient. Counterfactual explanations may resemble ordinary human explanations, but they did not consistently support calibrated trust.

Ersozlu et al. offer a recent applied example. They evaluate an AI-supported workflow for teacher performance assessment. The system generated rubric-linked feedback and cited evidence from portfolios. Assessors retained authority to accept, modify, or reject suggestions. The workflow reduced perceived workload and marking time. It also supported more structured feedback and moderation. The authors describe the AI as a "third eye" rather than an autonomous marker [5]. This design is promising because the explanation is tied to evidence and embedded in a professional checking process. However, the study used different submissions in the baseline and AI-supported conditions. It therefore supports practical feasibility more strongly than causal claims about explanation effects.

The meta-analysis by Schemmer et al. places these positive findings in context. Across eligible user studies, XAI-assisted decision-making performed better than no assistance. However, explanations did not produce a clear additional benefit over AI predictions alone [8]. This distinction is crucial. A study cannot show that explanations improve performance merely by comparing XAI assistance with unaided human judgment. The relevant test is whether the explanation adds value beyond the recommendation itself.

5 When Explanations Increase Trust Without Better Decisions

Zhang et al. show that confidence scores and explanations affect users differently. Confidence information improved case-specific trust calibration. Local explanations were less reliable as a decision aid. Even when users adjusted trust, joint accuracy did not necessarily rise. Human–AI collaboration improves only when the human can correct cases in which the AI is weak [12]. An explanation cannot create missing domain knowledge.

Shabankareh et al. examine perceived transparency and trust in healthcare applications. Their structural model shows no significant direct effect of perceived transparency on trust. Transparency influenced trust indirectly through perceived fairness, while perceived control was not a significant mediator. Privacy concerns strengthened the role of fairness for some users [9]. This study does not test decision accuracy. Its value lies elsewhere. It shows that trust may be shaped by social and ethical judgments rather than by a user's ability to evaluate a recommendation. Perceived transparency can therefore operate as a signal of fairness, not as proof of understanding.

Albashrawi's multidisciplinary review reaches a compatible conclusion at a broader level. GenAI can support information processing, efficiency, and personalization across healthcare, law, business, and education. Yet misinformation, bias, and weak transparency remain persistent limitations. The review stresses human oversight and responsible governance [2]. Because it covers 101 articles and many domains, it does not isolate explanation effects. It does, however, show why trust in fluent generative systems is risky. GenAI outputs can appear coherent even when they are unreliable.

The evidence suggests that explanations often affect perceived credibility more consistently than objective performance. This is not always harmful. Trust can support adoption and efficient collabo-

ration. The problem arises when trust is not calibrated to case-level accuracy. Explanations should therefore be evaluated by whether they help users discriminate correct from incorrect advice. General satisfaction and perceived clarity are insufficient.

6 The Harm of Misleading Explanations

The strongest negative evidence concerns explanations that are inaccurate, irrelevant, or difficult to verify. Cabitza et al. study correct AI advice paired with misleading explanations in logic tasks. They describe an "XAI halo effect." Participants were influenced by the explanation even though the advice itself was correct. Misleading explanations reduced their ability to evaluate relevance and sometimes caused them to reject useful advice [3]. The study reveals that advice quality and explanation quality must be evaluated separately.

Spitzer et al. examine the longer-term consequences of incorrect explanations. Their experiment included four conditions: no AI support, AI advice only, correct explanations, and incorrect explanations. Correct explanations produced the strongest collaborative performance. Incorrect explanations impaired team performance relative to correct explanations and damaged later procedural knowledge. Participants learned flawed distinctions from the AI and performed worse after support was removed [10]. This is more serious than immediate overreliance. A misleading explanation can alter a person's mental model and contaminate future independent judgment.

These studies expand the usual focus on incorrect AI recommendations. A correct recommendation can still be harmful when paired with a poor explanation. Conversely, an incorrect recommendation can become more persuasive when accompanied by a coherent story. The main danger is not opacity alone. It is the combination of opacity with plausible post-hoc justification.

Peters provides a conceptual reason for this effect. Human explanations normally imply a relation between stated reasons and future conduct. AI explanation modules may be disconnected from the prediction process. A user may nevertheless assume that the explanation expresses the system's real reasons. This encourages a level of trust that the technical architecture does not justify [7]. The resulting problem is not only low transparency. It is false transparency.

7 Moderating Factors and Human–AI Workflow

7.1 Expertise and intuition

User expertise is one of the most consistent moderators. Janssen et al. find that experienced participants make more correct decisions and identify more algorithmic errors. Wang and Yin find that explanations are less useful in unfamiliar domains. Chen et al. show how intuition can help users challenge AI. These findings suggest that explanations do not replace expertise. They are tools through which expertise may be applied.

This creates a tension for public-facing systems. Non-experts may need explanations most, but they may also be least able to assess them. A detailed feature chart can appear authoritative without giving a novice a valid standard for comparison. Systems should therefore support verification rather than only disclosure. Examples, counterevidence, limitations, and uncertainty may be more useful than a polished account of feature influence.

7.2 Timing and anchoring

Agudo et al. show that the timing of AI advice changes its influence. Participants judged criminal cases with correct or incorrect algorithmic support. Incorrect advice reduced accuracy most strongly when it was shown before participants formed their own judgment [1]. Early advice acts as an anchor. The human task becomes confirmation or adjustment rather than independent assessment.

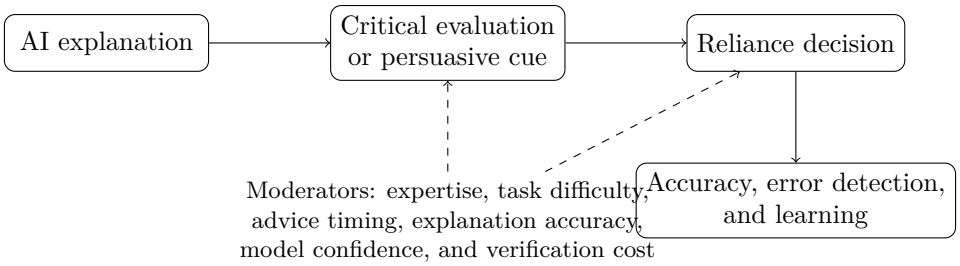


Figure 1: A synthesis of the reviewed evidence. Explanations influence decisions through user evaluation, but the pathway is moderated by the task and interface.

This finding is not an explanation study in the narrow sense. It is still central to XAI design. An explanation shown before independent reasoning may deepen the anchor by giving it a justification. A workflow that asks users to make an initial judgment first may preserve independent evidence processing. The sequence of interaction can therefore matter as much as the explanation format.

7.3 Accuracy, confidence, and uncertainty

Confidence information can help users estimate reliability, as Zhang et al. demonstrate. Yet confidence must be meaningful and understandable. A numerical probability may be misread. It may also be overvalued when the underlying model is poorly calibrated. Explanations should communicate uncertainty without turning it into another persuasion cue.

A useful design target is complementary performance. Let H denote unaided human accuracy, M model accuracy, and T team accuracy. The value of collaboration can be expressed as:

$$G = T - \max(H, M). \tag{3}$$

A positive G indicates that the human–AI team exceeds its strongest member. Several reviewed studies show that explanation can increase reliance without producing positive team gain. Chen et al.'s example-based condition is notable because it moved closer to complementarity. Schemmer et al.'s meta-analysis shows that such gains are not yet consistent across studies.

8 Methodological Weaknesses in the Literature

The literature is informative but fragmented. First, many studies use simplified classification tasks. These tasks provide clear ground truth, but they may not represent clinical, legal, or educational practice. Chen et al. acknowledge that their tasks were low stakes and did not require specialist expertise. Janssen et al. also note that their structured cases excluded ambiguous decisions. Applied studies improve realism, but they often sacrifice experimental control.

Second, trust is measured inconsistently. Some studies use self-report scales. Others infer reliance from advice acceptance. Some use final accuracy or calibration. These outcomes should not be combined without care. Shabankareh et al. study trust perceptions but not behavioural accuracy. Their findings should therefore not be treated as evidence that transparency improves decisions.

Third, explanation and advice are sometimes confounded. Schemmer et al. show why an AI-plus-explanation condition must be compared with the same AI advice without explanation. Otherwise, improvement may come from the recommendation rather than from XAI.

Fourth, many studies are short. Trust develops through repeated success, failure, and correction. Spitzer et al. are unusual in measuring post-collaboration performance. Their results indicate that immediate accuracy can hide damage to later learning.

Fifth, explanation quality is often assumed rather than independently tested. Cabitza et al. and

Spitzer et al. show that correctness and relevance are causal variables, not minor design details. Future studies should evaluate whether participants can verify explanations and whether the explanations are faithful to the model.

Finally, several studies rely on online samples or small groups. This limits generalization to professionals. Ersozlu et al. provide a useful professional workflow, but their design does not support strong causal inference. More field experiments are needed in which practitioners face realistic consequences and remain accountable for final decisions.

9 Implications for Research and Design

The reviewed literature supports five practical principles. First, systems should optimise appropriate reliance rather than trust. A design that increases trust across all cases is unsafe when the AI is fallible. Second, explanations should be evaluated against behavioural outcomes. Decision accuracy, error detection, and later independent performance are stronger tests than perceived clarity.

Third, explanation interfaces should make disagreement possible. Users need access to evidence, uncertainty, counterexamples, and limitations. They also need time to form an independent view. Agudo et al.'s findings suggest that delaying AI advice until after an initial judgment may reduce anchoring.

Fourth, explanation design should reflect user expertise. Experts may benefit from detailed evidence and model limitations. Non-experts may need simpler comparisons, warnings, or escalation to a human professional. More information is not always more transparency.

Fifth, explanations must be accurate and relevant. A plausible explanation is not enough. Systems should abstain when a reliable explanation cannot be generated. Cabitza et al. explicitly recommend caution when explanation quality is uncertain. Spitzer et al. show that this requirement protects not only the current decision but also human learning.

Future studies should compare explanation types within the same task, model, and sample. They should manipulate AI correctness and explanation correctness independently. They should also measure trust, reliance, accuracy, and learning as separate outcomes. Longer studies with professionals are especially important. The main research question should shift from "Do users like the explanation?" to "Can users use it to identify when the AI should not be followed?"

10 Conclusion

This review examined whether AI explanations improve human decisions or mainly increase perceived trust. The evidence does not support a universal answer. Explanations can improve performance. Yet the benefit appears only under clear conditions. The explanation must be accurate, relevant, and easy to check. The user must also have enough knowledge to question the recommendation. When these conditions are present, explanations can help users detect errors, combine human intuition with model output, and make more defensible decisions.

The positive findings are therefore important but limited. Janssen et al. show that transparency and experience can improve error detection. Chen et al. find that example-based explanations can support stronger human–AI performance than feature-based explanations. Ersozlu et al. also show the practical value of evidence-linked support in a professional workflow. These studies suggest that explanations work best when they direct attention to evidence and preserve human authority. They are less useful when they simply present a polished account of the model's output.

The negative findings are more consistent. Several studies show that explanations can increase reliance without improving accuracy. Confidence information may improve trust calibration, but calibration alone does not guarantee better team performance. Users still need knowledge that complements the AI. Perceived transparency can also increase trust through fairness or credibility

judgments rather than through genuine understanding. In such cases, trust rises because the system appears responsible, not because the user can identify when it is wrong.

Misleading explanations create the clearest risk. Cabitza et al. and Spitzer et al. show that poor explanations can damage both immediate decisions and later learning. A correct recommendation does not protect the user when its explanation is irrelevant or false. A plausible explanation can also make incorrect advice more convincing. This means that explanation quality must be assessed independently from prediction quality. An AI system should not provide an explanation merely because one can be generated.

The review also highlights the importance of workflow design. Expertise, task familiarity, timing, model confidence, and verification cost all shape the effect of an explanation. Agudo et al. show that early AI advice can anchor human judgment. Wang and Yin show that explanation methods lose value when users lack domain knowledge or when models become more complex. These findings suggest that interface design is not a secondary issue. It directly affects whether explanations support reflection or encourage passive acceptance.

The main conclusion is that explainable AI should not aim to make users trust AI more. It should help users decide when trust is justified. This requires a shift in evaluation. Studies should measure final accuracy, error detection, appropriate reliance, and later independent performance. Self-reported trust and perceived clarity remain useful, but they are not sufficient. Future research should use longer studies, professional participants, and realistic consequences. It should also test prediction correctness and explanation correctness as separate variables.

AI explanations are therefore neither a guaranteed safeguard nor a purely persuasive device. They are decision tools whose value depends on their design and use. A useful explanation strengthens human judgment. A poor explanation can weaken it. The most responsible objective is not transparency in appearance, but transparency that enables verification, disagreement, and accountable action.

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